

SketchSearch: Fine-tuning Reference Maps to Create Exercises In Support of Video-based Learning for Surgeons

Jingying Wang
The University of Michigan
Ann Arbor, Michigan, USA

Vitaliy Popov
The University of Michigan
Ann Arbor, Michigan, USA

Xu Wang
The University of Michigan
Ann Arbor, Michigan, USA

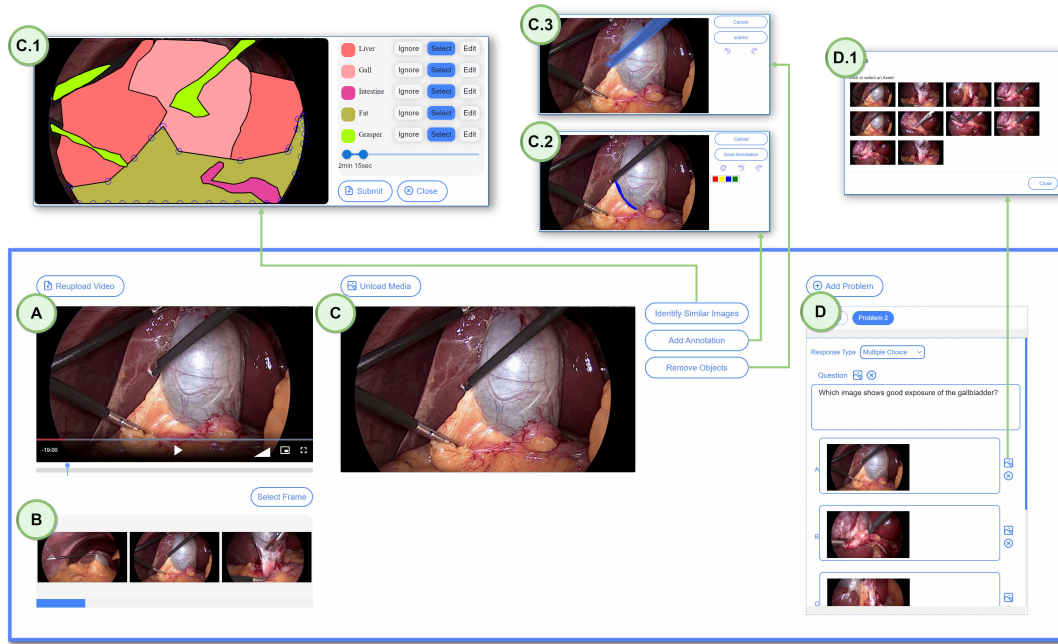


Figure 1: In the SketchSearch interface, the video panel (A) displays a laparoscopic cholecystectomy surgery video, alongside the keyframes (B) of the video. The image processing panel (C) offers three functionalities: identifying similar images (C.1), adding annotations to the image (C.2), and removing components from the image (C.3). The canvas for similar image identification (C.1) enables users to retrieve frames based on semantic references such as "more fat obstructing the view" through adjusting the sketch of the referenced image. Finally, the user can re-use the retrieved images to create exercises (D).

ABSTRACT

Video-based surgical coaching involves mentors reviewing surgery footage with trainees. Although effective, its use is sporadic due to time constraints. We propose AI-augmented coaching through SketchSearch, allowing experts to create exercises with automated feedback in surgery videos for self-learning. Surgeons often seek specific scenes for teaching, relying on visual cues. SketchSearch simplifies this through a three-step process: key frame extraction, template reference maps creation via image segmentation, and fine-tuning for frame retrieval.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST '23 Adjunct, October 29–November 01, 2023, San Francisco, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0096-5/23/10.

<https://doi.org/10.1145/3586182.3615816>

CCS CONCEPTS

• Human-centered computing → Web-based interaction.

KEYWORDS

video-based coaching, video navigation, keyframe extraction, surgical training, minimally invasive surgery

ACM Reference Format:

Jingying Wang, Vitaliy Popov, and Xu Wang. 2023. SketchSearch: Fine-tuning Reference Maps to Create Exercises In Support of Video-based Learning for Surgeons. In *The 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Adjunct)*, October 29–November 01, 2023, San Francisco, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3586182.3615816>

1 INTRODUCTION

Laparoscopic surgeries, involving small incisions and a camera-guided procedure, significantly rely on video streams, making video-based coaching an effective training method [8]. However, its usage



Figure 2: The SketchSearch platform provides a canvas for users to make fine-tuning on the sketch, enabling three modes for the components in the scene: 1) "Ignore" mode allows users to exclude the factor of a specific component during the searching process, 2) "Select" mode activates the object for moving, resizing, and rotating, 3) "Edit" mode allows users to drag the vertex of the polygons to adjust the shape and position of the component.

is limited due to implementation challenges, time commitments for coaches, and lack of effective postoperative feedback [2, 4, 9].

This work seeks to increase the accessibility and scalability of video-based coaching by streamlining the process. Recognizing the necessity of expert involvement in tracking subtle changes in surgical recordings, we aim to facilitate feedback provision. While existing online courses often use text-based quizzes [1, 5, 10–12], surgical training needs visual and positional cues, a need unmet by current superficial image-based question methods.

Addressing surgeons' difficulties in visually identifying teachable moments and locating specific video frames for educational exercises, we propose SketchSearch, a system that uses AI to assist surgeons in retrieving images with interests for exercises. SketchSearch segments a frame in a surgery recording into components, generates a reference map, and lets the user adjust component shapes to indicate retrieval goals. The system then finds video frames most similar to the user-modified reference map. We introduce semantics into image retrieval via segmentation, allowing more user flexibility in specifying needs, like position or exposure of a structure.

2 SKETCHSEARCH INTERFACE

SketchSearch enables surgeons to quickly locate episodes of interest and find frames that fulfill specific requirements already in their minds, such as "more exposure of the gallbladder," "more fat obstructing the view," or "the surgical tool placed between gallbladder and fat." As shown in Fig.1, the interface comprises three main sections: 1) Raw Material Retrieval Section (A) & (B) which display a video and a menu of keyframes for the video. Users can either select a frame or open a keyframe for further processing. 2) Image Processing Panel (C) where, users can identify similar images to the keyframe (C.1) or add annotations (C.2) and remove objects from any frames (C.3). Once they have finished, they can proceed

to the next section. 3) Question Form (D) where users can insert the processed images (D.1) into the question form and save it for future use.

2.1 Fine-tuning Sketch Searching (C.1)

When a user selects "Identify Similar Image" for a keyframe, an initial sketch template is generated, containing movable, resizable, and rotatable polygons representing different scene components, as shown in Fig.2. Users can disregard specific components during the search by hiding them with the "ignore" button. If they want a component included, they can activate it with the "select" button and modify its position and size.

After editing the canvas, users can define the search range or use the default setting. The system then returns nine suggested frames based on the specifications. Users can add these suggestions to their collection, and by hovering over the images, they can see a value indicating the similarity between the frame and the reference sketch, with a lower value indicating higher alignment.

2.2 Algorithms

In order to reduce the cognitive burden associated with generating sketches from scratch, we present a set of templates derived from video frames for users to edit with minimal effort. We introduce a novel computational pipeline tailored for the analysis of laparoscopic videos, which leverages neural feature extraction for keyframe extraction and employs scene segmentation to identify frames containing specific structures.

2.2.1 Video Keyframe Extraction. Our algorithm determines representative frames in laparoscopic cholecystectomy videos by detecting the centroids of similar visual features. This process involves extracting features from each frame using pretrained deep neural networks like VGG19's FC-4096 layer, known for its capacity to

capture content and texture details. Next, we calculate the cosine similarity between the features of all frame pairs, generating a similarity score between 0 (completely dissimilar) and 1 (identical). By visualizing this similarity matrix, as shown in Fig.3a, we target the centers of the light patches which represent segments with visually consistent and similar frames. Through applying a mean filter along the diagonal axis of the matrix, it produces a 1-D signal that reflects local average similarity as shown in Fig.3b. The peaks of this signal correspond to the patch centers.

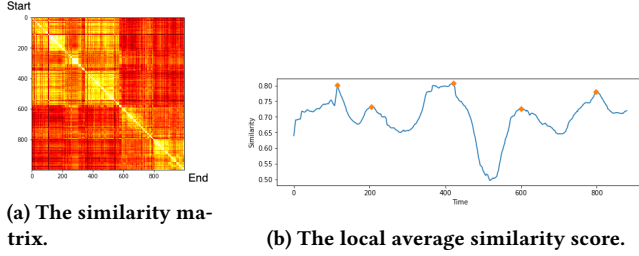


Figure 3: The visual similarity matrix is first computed for the video. Then, a square-shaped mean filter slide along the diagonal axis to compute the local similarity at each time stamp.

2.2.2 Fine-tuning-sketch-based Searching. The second part of our algorithm involves matching specific structures, modified by users based on templates, to the frames in the video.

Laparoscopic image segmentation. To comprehend the structural composition of laparoscopic images, we first apply scene segmentation models to analyze the distribution of various components within the images. We utilize a U-Net [7], trained on the CholecSeg8k dataset [3] and m2caiSeg [6], by considering the mutually labeled components. The dataset is divided into 80% training data and 20% test data, achieving a mean Dice score of 0.75.

Template Conversion. After identifying all pixels on the canvas, we determine the alpha shape to identify the contours of the components, and stack the component polygons on the canvas in (C.1) with the following order: "Abdominal Wall", "Liver", "Gallbladder", "Fat", "Intestine", "Blood", and "Tools".

Searching algorithm. The central concept of the searching algorithm is to identify frames whose segmentation maps are closest to the templates, as determined by the Mean Squared Error (MSE).

Our system enables users to modify templates as required. Upon completion, a reference map M_r is generated, so that we can compare M_r with the segmentation map M_i of each frame i for the one with the highest similarity.

To further emphasize the changes made by users to the templates, we incorporate a weighted measurement of the MSE between the reference map and the frames' segmentation maps,

$$D_i = ||W \circ (M_r - M_i)||_2, \quad (1)$$

where

$$W(x, y) = \begin{cases} 2, & \text{if pixel } (x, y) \text{ is in the region of modification,} \\ 1, & \text{otherwise.} \end{cases} \quad (2)$$

The frame recommended to users is then determined by:

$$\operatorname{argmin}_i D_i. \quad (3)$$

REFERENCES

- [1] Bidyut Das, Mukta Majumder, Santanu Phadikar, and Arif Ahmed Sekh. 2021. Automatic question generation and answer assessment: a survey. *Research and Practice in Technology Enhanced Learning* 16, 1 (18 Mar 2021), 5. <https://doi.org/10.1186/s41039-021-00151-1>
- [2] Andrew C Esposito, Nathan A Coppersmith, Erin M White, and Peter S Yoo. 2022. Video coaching in surgical education: utility, opportunities, and barriers to implementation. *Journal of Surgical Education* 79, 3 (2022), 717–724.
- [3] W. Y. Hong, C. L. Kao, Y. H. Kuo, J. R. Wang, W. L. Chang, and C. S. Shih. 2020. CholecSeg8k: A Semantic Segmentation Dataset for Laparoscopic Cholecystectomy Based on Cholec80. arXiv:2012.12453 [cs.CV]
- [4] Deborah S Keller, Emily R Winslow, Joel E Goldberg, and Vanita Ahuja. 2021. Video-based coaching: current status and role in surgical practice (Part 1) from the society for surgery of the alimentary tract, health care quality and outcomes committee. *Journal of Gastrointestinal Surgery* 25, 9 (2021), 2439–2446.
- [5] Ghader Kurdi, Jared Leo, Bijan Parsia, Uli Sattler, and Salam Al-Emari. 2020. A Systematic Review of Automatic Question Generation for Educational Purposes. *International Journal of Artificial Intelligence in Education* 30, 1 (01 Mar 2020), 121–204. <https://doi.org/10.1007/s40593-019-00186-y>
- [6] Salman Maqbool, Aqsa Riaz, Hasan Sajid, and Osman Hasan. 2020. m2caiSeg: Semantic Segmentation of Laparoscopic Images using Convolutional Neural Networks. arXiv:2008.10134 [cs.CV]
- [7] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. arXiv:1505.04597 [cs.CV]
- [8] Pritam Singh, Rajesh Aggarwal, Muaaz Tahir, Philip H. Pucher, and Ara Darzi. 2015. A Randomized Controlled Study to Evaluate the Role of Video-based Coaching in Training Laparoscopic Skills. *Annals of Surgery* 261, 5 (2015). https://journals.lww.com/annalsofsurgery/Fulltext/2015/05000/A_Randomized_Controlled_Study_to_Evaluate_the_Role.9.aspx
- [9] Rebecca A. Snyder, Margaret J. Tarpley, John L. Tarpley, Mario Davidson, Colleen Brophy, and Jeffery B. Dattilo. 2012. Teaching in the Operating Room: Results of a National Survey. *Journal of Surgical Education* 69, 5 (2012), 643–649. <https://doi.org/10.1016/j.jsurg.2012.06.007>
- [10] Xu Wang, Srinivasa Teja Talluri, Carolyn Rose, and Kenneth Koedinger. 2019. UpGrade: Sourcing Student Open-Ended Solutions to Create Scalable Learning Opportunities. In *Proceedings of the Sixth (2019) ACM Conference on Learning @ Scale* (Chicago, IL, USA) (L@S '19). Association for Computing Machinery, New York, NY, USA, Article 17, 10 pages. <https://doi.org/10.1145/3330430.3333614>
- [11] Zichao Wang, Jakob Valdez, Debshila Basu Mallick, and Richard G. Baraniuk. 2022. Towards Human-Like Educational Question Generation with Large Language Models. In *Artificial Intelligence in Education*, Maria Mercedes Rodrigo, Noburu Matsuda, Alexandra I. Cristea, and Vania Dimitrova (Eds.). Springer International Publishing, Cham, 153–166.
- [12] Iman Yekkehzaare, Tirdad Barghi, and Paul Resnick. 2020. QMaps: Engaging Students in Voluntary Question Generation and Linking. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376882>